

Table 6. Independent solver validation

Panel A. Independent solver validation

Observable	Worst absolute error (dimensionless)	Worst unit-scale-normalized error	Fixed normalization scale	Near zero policy	Absolute tolerance (dimensionless)	Passes	Reference solver	Independent solver	Scope
density_matrix	6.449e-13	6.449e-13	1	fixed unit scale; no data-derived denominator	2.000e-09	Yes	constant-generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
doublet_population	1.270e-12	1.270e-12	1	fixed unit scale; no data-derived denominator	3.000e-09	Yes	constant-generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
quartet_population	1.239e-12	1.239e-12	1	fixed unit scale; no data-derived denominator	3.000e-09	Yes	constant-generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
survival	5.997e-13	5.997e-13	1	fixed unit scale; no data-derived denominator	3.000e-09	Yes	constant-generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
doublet_reaction_yield	5.284e-13	5.284e-13	1	fixed unit scale; no data-derived denominator	3.000e-09	Yes	constant-generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
quartet_reaction_yield	7.977e-14	7.977e-14	1	fixed unit scale; no data-derived denominator	3.000e-09	Yes	constant-generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation

Worst discrepancies compare the matrix exponential with the adaptive Dormand-Prince solver. The normalized metric divides by the fixed unit scale for bounded dimensionless quantities, never by a near-zero result. Presentation values are rounded. All 10 source rows and all 16 source columns remain in table06_independent_solver_validation.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 6. Independent solver validation

Panel A. Independent solver validation (continued)

Observabl e	Worst absolute error (dimens ionless)	Worst unit- scale- normalized error	Fixed nor malizatio n scale	Near zero policy	Absolute tolerance (d imensionless)	Passes	Reference solver	Independent solver	Scope
primary_su peroxide_y ield	4.529e-13	4.529e-13	1	fixed unit scale; no data- derived denominator	3.000e-09	Yes	constant- generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
escape_yie ld	2.127e-13	2.127e-13	1	fixed unit scale; no data- derived denominator	3.000e-09	Yes	constant- generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
reference_ probabilit y_accounti ng	4.885e-15	4.885e-15	1	fixed unit scale; no data- derived denominator	1.000e-08	Yes	constant- generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation
independen t_probabil ity_accoun ting	6.661e-16	6.661e-16	1	fixed unit scale; no data- derived denominator	1.000e-08	Yes	constant- generator matrix exponential Pade(13)	independent adaptive Dormand-Prince RK5(4)	numerical implementation agreement, not physical validation

Worst discrepancies compare the matrix exponential with the adaptive Dormand-Prince solver. The normalized metric divides by the fixed unit scale for bounded dimensionless quantities, never by a near-zero result. Presentation values are rounded. All 10 source rows and all 16 source columns remain in table06_independent_solver_validation.csv; 6 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.